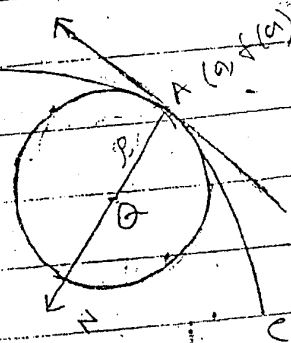


7.8-1

EX. 7.8

The centre of curvature to the curve C defined by $y = f(x)$ at $A = (a, f(a))$ is the pt Q on the normal to the curve at $A(a, f(a))$ on concave side of curve s.t. $|AQ| = \rho$



The circle of curvature of C at $(a, f(a))$ is the circle with centre Q and radius ρ

Circle of Curvature is also called OSCULATING CIRCLE

EVOLUTE is the locus of centres of curvature of a curve and the curve is called Involute of this locus of centres of curvature.



The Normals to a curve are Tangents to its Evolute

Coordinates (α, β) of the centre of Curvature are

$$\alpha = x - \frac{y' [1 + y'^2]}{y''}$$

$$\beta = y + \frac{[1 + y'^2]}{y''}$$

$$\rho = \frac{[1 + (y')^2]^{3/2}}{|y''|}$$

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Eg of Osculating Circle $(x - \alpha)^2 + (y - \beta)^2 = \rho^2$

7.8-2

Ex 7.8

Eq of Osculating Circle

$$(x-\alpha)^2 + (y-\beta)^2 = \rho^2$$

① $y = \ln x$ at $(1, 0)$

$$y' = \frac{1}{x}$$

$$y'' = -\frac{1}{x^2}$$

$$y'(1,0) = \frac{1}{1} = 1$$

$$y''(1,0) = -\frac{1}{1} = -1$$

$$\rho = \frac{[1+(y')^2]^{3/2}}{|y''|}$$

$$= \frac{[1+(\frac{1}{x})^2]^{3/2}}{|-\frac{1}{x^2}|}$$

$$= \frac{[x^2+1]^{3/2}}{\frac{1}{x^2}}$$

$$\rho_{at(1,0)} = \frac{[1+(y'(1,0))^2]^{3/2}}{|y''(1,0)|}$$

$$= \frac{(x^2+1)^{3/2}}{x^3} \cdot \frac{x^2}{1}$$

$$= \frac{(x^2+1)^{3/2}}{x}$$

$$\rho_{at(1,0)} = \frac{(1+1)^{3/2}}{1} = 2^{3/2} = 2\sqrt{2}$$

$$= 2\sqrt{2}$$

Centre of Curvature (α, β) is

$$\alpha = x - \frac{y'(1+y'^2)}{y''}$$

$$\alpha_{(1,0)} = 1 - \frac{1(1+1)}{-1} = 1 + 2 = 3$$

$$\beta = y + \frac{(1+y'^2)}{y''}$$

$$\beta_{(1,0)} = 0 + \frac{(1+1)}{-1} = -2$$

Eq of Osculating Circle (circle of curvature) is $(x-\alpha)^2 + (y-\beta)^2 = \rho^2$

$$\therefore (x-3)^2 + (y+2)^2 = (2\sqrt{2})^2 \Rightarrow (x-3)^2 + (y+2)^2 = 8$$

7.8-3

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② $\frac{x^2}{4} - \frac{y^2}{9} = 1$ at $(-2, 0)$

$$\frac{2x}{4} - \frac{2yy'}{9} = 0$$

$$\frac{x}{2} - \frac{2yy'}{9} = 0$$

$$\frac{9x - 4yy'}{18} = 0 \Rightarrow 9x - 4yy' = 0 \Rightarrow \frac{9x}{4y} = y'$$

$$y'' = \frac{9}{4} \left(\frac{y(1) - x y'}{y^2} \right)$$

$$= \frac{9}{4} \left(\frac{y - x \frac{9x}{4y}}{y^2} \right) = \frac{9}{4} \frac{(4y^2 - 9x^2)}{4y^3} = \frac{9(4y^2 - 9x^2)}{16y^3}$$

$$\rho = \frac{|1 + y'^2|^{3/2}}{|y''|} = \frac{\left(1 + \left(\frac{9x}{4y}\right)^2\right)^{3/2}}{\left|\frac{9(4y^2 - 9x^2)}{16y^3}\right|} = \frac{\left((4y)^2 + (9x)^2\right)^{3/2}}{|(4y)^{3/2} 9(4y^2 - 9x^2)|}$$

$$\rho = \frac{(16y^2 + 81x^2)^{3/2}}{|4 \cdot 64y^3 \cdot 9(4y^2 - 9x^2)|} = \frac{(16y^2 + 81x^2)^{3/2}}{|36(4y^2 - 9x^2)|}$$

$$\rho \text{ at } (-2, 0) = \frac{|16(0)^2 + 81(-2)^2|^{3/2}}{|36(4(0) - 9(-2)^2)|} = \frac{|81(4)|^{3/2}}{|36(-36)|} = \frac{|9^2 \cdot 2|^{3/2}}{|36(-36)|}$$

$$= \frac{9^3 \cdot 2^3}{|36(-36)|} = \frac{9(8)}{|4(-4)|} = \frac{9(2)}{|(-1)|} = \frac{9}{|-2|} = \frac{9}{2}$$

check it

find y' at $(-2, 0)$

y'' at $(-2, 0)$

ρ at $(-2, 0)$ using

α at $(-2, 0)$ using

β at $(-2, 0)$ using

Centre of Curvature (α, β) is

$$\alpha = x - \frac{y'(1+y'^2)}{y''} = x - \frac{9x}{4y} \left(1 + \frac{81x^2}{16y^2}\right) \cdot \frac{9(4y^2 - 9x^2)}{16y^3}$$

$$\alpha = x - \frac{9x(16y^2 + 81x^2)}{4y \cdot 16y^2} \cdot \frac{9(4y^2 - 9x^2)}{16y^3} = x - \frac{9x(16y^2 + 81x^2)}{4 \cdot 64y^3} \cdot \frac{9(4y^2 - 9x^2)}{16y^3}$$

$$\alpha = x - \frac{9x(16y^2 + 81x^2)}{36(4y^2 - 9x^2)}$$

$$\alpha(-2, 0) = (-2) - \frac{9(-2)(0 + 81(-2)^2)}{36(0 - 9(-2)^2)} = (-2) + \frac{18[81(4)]}{(36)(-36)}$$

$$\frac{(-2) + \frac{81(4)}{2(-36)}}{-2 + \frac{81(2)}{-36}} = \frac{-2 + \frac{81}{-18}}{-2 + \frac{9}{-2}} = \frac{+4+9}{-2} = \frac{13}{-2}$$

$$\beta = y + \frac{(1+y'^2)}{y''} = y + \frac{(1 + \frac{81x^2}{16y^2})}{\frac{9(4y^2 - 9x^2)}{16y^3}} = y + \frac{(16y^2 + 81x^2)}{16y^2} \cdot \frac{16y^3}{9(4y^2 - 9x^2)}$$

$$= y + \frac{(16y^2 + 81x^2)}{9(4y^2 - 9x^2)}$$

$$\beta(-2, 0) = 0 - 0 = 0$$

7.8-5

Eq of the Osculating circle is

$$(x-\alpha)^2 + (y-\beta)^2 = r^2$$

$$\left\{x - \left(-\frac{13}{2}\right)\right\}^2 + \{y - 0\}^2 = \left(\frac{9}{2}\right)^2$$

$$\left(x + \frac{13}{2}\right)^2 + y^2 = \frac{81}{4}$$

③ $x = a \cos \theta$

$y = b \sin \theta$

$$\frac{dx}{d\theta} = -a \sin \theta$$

$$\frac{dy}{d\theta} = b \cos \theta$$

$$\gamma' = \frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx} = \frac{b \cos \theta}{-a \sin \theta} = -\frac{b}{a} \cot \theta$$

$$\gamma'' = \frac{d^2 y}{dx^2} = \frac{b}{a} \left[\frac{-\operatorname{Cosec}^2 \theta \frac{d\theta}{dx}}{\frac{1}{a \sin \theta}} \right] = -\frac{b}{a} \left(\frac{\operatorname{Cosec}^2 \theta}{\sin \theta} \right)$$

$$\gamma'' = \frac{-b}{a^2} \frac{\operatorname{Cosec}^2 \theta}{\sin \theta} = \frac{-b}{a^2 \sin^3 \theta}$$

$$\alpha = x - \frac{\gamma' (1 + (\gamma')^2)}{\gamma''}$$

$Q(\alpha, \beta) \equiv \text{Centre of Curvature}$

$$\alpha = a \cos \theta - \left(\frac{-\frac{b}{a} \cot \theta}{\frac{-b}{a^2 \sin^3 \theta}} \right) \left(1 + \frac{b^2 \cot^2 \theta}{a^2} \right)$$

7.8-6

$$\alpha = a \cos \theta - \frac{\cot \theta (a^2 + b^2 \cot^2 \theta)}{a} \sqrt{\frac{a^2 \sin^3 \theta}{a^2}}$$

$$= \frac{a^2 \cos \theta - \cot \theta (a^2 + b^2 \cot^2 \theta) \sin^3 \theta}{a}$$

$$= \frac{1}{a} (a^2 \cos \theta - (\cot \theta a^2 + b^2 \cot^3 \theta) \sin^3 \theta)$$

$$= \frac{1}{a} \left(a^2 \cos \theta - \left(\frac{\cos \theta}{\sin \theta} a^2 \sin^3 \theta + b^2 \frac{\cos^3 \theta}{\sin^3 \theta} \sin^3 \theta \right) \right)$$

$$= \frac{1}{a} (a^2 \cos \theta - a^2 \cos \theta \sin^2 \theta - b^2 \cos^3 \theta)$$

$$= \frac{1}{a} (a^2 \cos \theta (1 - \sin^2 \theta) - b^2 \cos^3 \theta)$$

$$\alpha = \frac{1}{a} (a^2 \cos^3 \theta - b^2 \cos^3 \theta) = \frac{1}{a} \cos^3 \theta (a^2 - b^2)$$

$$a \alpha = \cos^3 \theta (a^2 - b^2) \quad (1)$$

$$\beta = \gamma + \frac{1 + (\gamma')^2}{\gamma''}$$

$$= b \sin \theta + \frac{1 + \left(-\frac{b}{a} \cot \theta \right)^2}{-\frac{b}{a^2 \sin^3 \theta}}$$

$$= b \sin \theta + \left(\frac{a^2 + b^2 \cot^2 \theta}{a^2} \right) \left(\frac{a^2 \sin^3 \theta}{-b} \right)$$

7.8-7

$$= b \sin \theta - \frac{1}{b} \left[\frac{a^2 \sin^3 \theta + b^2 \sin^3 \theta \cos^2 \theta}{\sin^2 \theta} \right]$$

$$= b \sin \theta - \frac{1}{b} (a^2 \sin \theta + b^2 \sin \theta \cos^2 \theta)$$

$$= b^2 \sin \theta - \frac{a^2 \sin^3 \theta + b^2 \sin \theta \cos^2 \theta}{b}$$

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$$b\beta = b^2 \sin \theta (1 - \cos^2 \theta) - a^2 \sin^3 \theta$$

$$b\beta = b^2 \sin^3 \theta - a^2 \sin^3 \theta = (b^2 - a^2) \sin^3 \theta \quad (1)$$

$$b\beta = (a^2 - b^2)(-\sin^3 \theta) \quad (11)$$

For Evolute we eliminate θ from (1) & (11)

Taking power $\frac{2}{3}$ on both sides of (1) & (11) & adding

$$(a^{\frac{2}{3}})^{\frac{2}{3}} + (b\beta)^{\frac{2}{3}} = (a^2 - b^2)^{\frac{2}{3}} (\cos^3 \theta)^{\frac{2}{3}} + (a^2 - b^2)^{\frac{2}{3}} (-\sin^3 \theta)^{\frac{2}{3}}$$

$$= (a^2 - b^2)^{\frac{2}{3}} \left[(\cos \theta)^{\frac{2}{3}} + (-\sin \theta)^{\frac{2}{3}} \right]$$

$$= (a^2 - b^2)^{\frac{2}{3}} \left[(\cos \theta)^2 + (-\sin \theta)^2 \right]$$

$$(a^{\frac{2}{3}})^{\frac{2}{3}} + (b\beta)^{\frac{2}{3}} = (a^2 - b^2)^{\frac{2}{3}} (\cos^2 \theta + \sin^2 \theta)$$

Put $\alpha = x$, $\beta = y$

$$(ax)^{\frac{2}{3}} + (by)^{\frac{2}{3}} = (a^2 - b^2)^{\frac{2}{3}} \quad \text{proved.}$$

7.8-8

Q4 $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

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$x = a \sec \theta$

$y = b \tan \theta$

$\frac{dx}{d\theta} = a \sec \theta \tan \theta$

$\frac{dy}{d\theta} = b \sec^2 \theta$

$y' = \frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx} = \frac{b \sec^2 \theta}{a \sec \theta \tan \theta} = \frac{b \sec \theta}{a \tan \theta}$

$y' = \frac{b}{a} \cdot \frac{1}{\cos \theta} \cdot \frac{\sin \theta}{\cos \theta} = \frac{b}{a \sin \theta} = \frac{b \operatorname{cosec} \theta}{a}$

$y'' = \frac{d^2 y}{dx^2} = \frac{b}{a} \left[-\operatorname{cosec} \theta \cot \theta \cdot \frac{d\theta}{dx} \right] = \frac{b}{a} \left[\frac{-\operatorname{cosec} \theta \cot \theta}{a \sec \theta \tan \theta} \right]$

$y'' = -\frac{b}{a^2} \left[\frac{1}{\sin \theta} \cdot \frac{\cos \theta}{\sin \theta} \cdot \frac{\cos \theta}{\sin \theta} \right] = -\frac{b \cos^2 \theta}{a^2 \sin^3 \theta}$

$Q(x, y)$ be centre of curvature

$x = x - \frac{y'(1+y'^2)}{y''} = a \sec \theta - \frac{b \operatorname{cosec} \theta \left(1 + \frac{b^2 \operatorname{cosec}^2 \theta}{a^2} \right)}{-\frac{b \cos^2 \theta}{a^2 \sin^3 \theta}}$

$= a \sec \theta - \frac{b \operatorname{cosec} \theta (a^2 + b^2 \operatorname{cosec}^2 \theta)}{a^2} \cdot \frac{a^2 \sin^3 \theta}{(-b) \cos^3 \theta}$

$= a \sec \theta + \frac{1}{a \sin \theta \cos^3 \theta} \cdot \frac{\sin^3 \theta (a^2 + b^2)}{\sin^2 \theta}$

$x = \frac{a}{\cos \theta} + \frac{\sin^2 \theta (a^2 \sin^2 \theta + b^2)}{a \cos^3 \theta \sin^2 \theta} = \frac{a^2 \cos^2 \theta + a^2 \sin^2 \theta + b^2}{a \cos^3 \theta}$

7.8-9

$$\alpha = \frac{(a^2 + b^2)}{a \cos^3 \theta}$$

$$\therefore \sin^2 \theta + \cos^2 \theta = 1$$

$$\alpha = \frac{(a^2 + b^2) \sec^3 \theta}{a}$$

$$\Rightarrow a\alpha = (a^2 + b^2) \sec^3 \theta \quad \text{--- (I)}$$

$$\beta = \gamma + \frac{(1 + \gamma'^2)}{\gamma''} = b \tan \theta + \left(1 + \frac{b^2 \cos^2 \theta}{a^2}\right) \frac{-\frac{b}{a^2} \cos^3 \theta}{\sin^3 \theta}$$

$$= b \tan \theta - \frac{a^2 \sin^3 \theta (a^2 + b^2 \cos^2 \theta)}{b \cos^3 \theta}$$

$$= \frac{b \sin \theta}{\cos \theta} - \frac{a^2 \sin^3 \theta}{b \cos^3 \theta} - \frac{b^2 \sin^3 \theta}{b \cos^3 \theta} \cdot \frac{1}{\sin^2 \theta}$$

$$= \frac{b^2 \cos^2 \theta \sin \theta - a^2 \sin^3 \theta - b^2 \sin \theta}{b \cos^3 \theta} = \frac{b^2 \sin \theta (\cos^2 \theta - 1) - a^2 \sin^3 \theta}{b \cos^3 \theta}$$

$$= \frac{b^2 \sin \theta (-\sin^2 \theta - a^2 \sin^3 \theta)}{b \cos^3 \theta} = \frac{-\sin^3 \theta (b^2 + a^2)}{b \cos^3 \theta}$$

$$\beta = -\frac{\tan^3 \theta (b^2 + a^2)}{b} \Rightarrow b\beta = -\tan^3 \theta (b^2 + a^2) \quad \text{--- (II)}$$

Taking power $\frac{2}{3}$ of (I) & (II) then subtracting (To eliminate θ)

$$(a\alpha)^{\frac{2}{3}} - (b\beta)^{\frac{2}{3}} = (a^2 + b^2)^{\frac{2}{3}} (\sec^3 \theta)^{\frac{2}{3}} - (a^2 + b^2)^{\frac{2}{3}} (-\tan^3 \theta)^{\frac{2}{3}}$$

$$= (a^2 + b^2)^{\frac{2}{3}} [\sec^2 \theta - \tan^2 \theta]$$

$$\therefore \sec^2 \theta - \tan^2 \theta = 1$$

Repla α by 'x' & β by 'y' $(ax)^{\frac{2}{3}} - (by)^{\frac{2}{3}} = (a^2 + b^2)^{\frac{2}{3}}$ proved

7.8-10

Q 5 $2xy = a^2$

$$y = \frac{a^2}{2x} \quad \text{--- (1)}$$

$$y' = \frac{dy}{dx} = -\frac{a^2}{2x^2}$$

$$y'' = \frac{d}{dx} \left(-\frac{a^2}{2x^2} \right) = \frac{a^2}{x^3}$$

Now $\alpha = x - \frac{y'(1+y')}{y''}$

$$\frac{1+y'^2}{y''} = \frac{1 + \left(-\frac{a^2}{2x^2} \right)^2}{\frac{a^2}{x^3}}$$

$$= \left(1 + \frac{a^4}{4x^4} \right) \left(\frac{x^3}{a^2} \right)$$

$$\frac{1+y'^2}{y''} = \left(\frac{4x^4 + a^4}{4x^4} \right) \left(\frac{x^3}{a^2} \right) = \left(\frac{4x^4 + a^4}{4a^2x} \right)$$

using (1) $\alpha = x - \left(\frac{a^2}{2x^2} \right) \left(\frac{4x^4 + a^4}{4a^2x} \right)$

$$\frac{1+y'^2}{y''} = \frac{4x^4 + a^4}{4a^2x} \quad \text{--- (II)}$$

$$\alpha = x + \frac{4x^4 + a^4}{8x^3}$$

$$= \frac{8x^4 + 4x^4 + a^4}{8x^3} = \frac{12x^4 + a^4}{8x^3} \quad \text{--- (III)}$$

$$\beta = y + \frac{(1+y'^2)}{y''}$$

using (1) $\beta = \frac{a^2}{2x} + \frac{4x^4 + a^4}{4a^2x} = \frac{2a^2(a^2) + 4x^4 + a^4}{4a^2x}$

$$\beta = \frac{3a^4 + 4x^4}{4a^2x} \quad \text{--- (IV)}$$

$$a(1 - \cos \theta)^2$$

7-8-11

For the evolute we eliminate x from (iii) & (iv)

$$\alpha + \beta = \frac{12x^4 + a^4}{8x^3} + \frac{3a^4 + 4x^4}{4a^2x}$$

$$= \frac{12x^4a + a^6 + 6x^4a + 8x^6}{8a^2x^3}$$

$$\alpha + \beta = \frac{(2x^2 + a^2)^3}{8a^2x^3}$$

$$\begin{aligned} (a+b)^3 &= a^3 + b^3 + 3ab(a+b) \\ (2x^2 + a^2)^3 &= (2x^2)^3 + (a^2)^3 + 3(2x^2)(a^2)(2x^2 + a^2) \\ &= 8x^6 + a^6 + 12x^4a + 6x^2a^4 \end{aligned}$$

$$(\alpha + \beta)^{\frac{2}{3}} = \frac{\left[(2x^2 + a^2)^3 \right]^{\frac{2}{3}}}{\left(\frac{8x^2a^3}{2a^{\frac{4}{3}}x} \right)^{\frac{2}{3}}} = \frac{(2x^2 + a^2)^2}{2a^{\frac{4}{3}}x}$$

$$(\alpha + \beta)^{\frac{2}{3}} = \frac{(2x^2 + a^2)^2}{4x^2a^{\frac{4}{3}}} \quad (v)$$

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$$\alpha - \beta = \frac{12x^4 + a^4}{8x^3} - \frac{3a^4 + 4x^4}{4a^2x} = \frac{12x^4a + a^6 - 6x^4a - 8x^6}{8a^2x^3}$$

$$\alpha - \beta = \frac{(a^2 - 2x^2)^3}{8a^2x^3} \quad (vi)$$

$$(\alpha - \beta)^{\frac{2}{3}} = \frac{\left[(a^2 - 2x^2)^3 \right]^{\frac{2}{3}}}{\left(\frac{8x^2a^3}{2a^{\frac{4}{3}}x} \right)^{\frac{2}{3}}} = \frac{(a^2 - 2x^2)^2}{2a^{\frac{4}{3}}x} = \frac{(a^2 - 2x^2)^2}{4x^2a^{\frac{4}{3}}} \quad (vii)$$

$$\begin{aligned} (\alpha + \beta)^{\frac{2}{3}} - (\alpha - \beta)^{\frac{2}{3}} &= \frac{(2x^2 + a^2)^2}{4x^2a^{\frac{4}{3}}} - \frac{(a^2 - 2x^2)^2}{4x^2a^{\frac{4}{3}}} \\ &= \frac{4x^4 + a^4 + 4x^2a^2 - a^4 - 4x^2a^2 + 4x^4}{4x^2a^{\frac{4}{3}}} \end{aligned}$$

7.8-12

$$(x+\beta)^{\frac{2}{3}} - (x-\beta)^{\frac{2}{3}} = \frac{8x^{\frac{2}{3}}\beta^{\frac{2}{3}}}{4x^{\frac{2}{3}}\beta^{\frac{2}{3}}} = 2\beta^{\frac{2}{3}}$$

$$(x+\beta)^{\frac{2}{3}} - (x-\beta)^{\frac{2}{3}} = 2\beta^{\frac{2}{3}}$$

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Put $x = \alpha$ $y = \beta$

$$(x+y)^{\frac{2}{3}} - (x-y)^{\frac{2}{3}} = 2y^{\frac{2}{3}}$$

(6) $x = a(1 - \sin \theta)$ $y = a(1 - \cos \theta)$

$$\frac{dx}{d\theta} = a(-\cos \theta) \quad \frac{dy}{d\theta} = a(\sin \theta)$$

$$y' = \frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx} = \frac{a(\sin \theta)}{a(-\cos \theta)} = \frac{\sin \theta}{1 - \cos \theta}$$

$$y'' = \frac{d^2y}{dx^2} = \left(\frac{(1 - \cos \theta) \cos \theta - \sin \theta (\sin \theta)}{(1 - \cos \theta)^2} \right) \frac{d\theta}{dx}$$

$$= \left(\frac{\cos \theta - \cos^2 \theta - \sin^2 \theta}{(1 - \cos \theta)^2} \right) \frac{1}{a(1 - \cos \theta)} = \frac{\cos \theta - (\cos^2 \theta + \sin^2 \theta)}{a(1 - \cos \theta)^3}$$

$$y'' = \frac{\cos \theta - 1}{a(1 - \cos \theta)^3} = -\frac{(1 - \cos \theta)}{a(1 - \cos \theta)^3} = -\frac{1}{a(1 - \cos \theta)^2}$$

$$\alpha = x - y' \left(\frac{1 + y'^2}{y''} \right) = a(1 - \sin \theta) - \left(\frac{\sin \theta}{1 - \cos \theta} \right) \left(\frac{1 + \frac{\sin^2 \theta}{(1 - \cos \theta)^2}}{-\frac{1}{a(1 - \cos \theta)^2}} \right)$$

7.8-13

$$= a(\theta - \sin \theta) + \frac{\sin \theta}{(1 - \cos \theta)} \left(\frac{(1 - \cos \theta)^2 + \sin^2 \theta}{(1 - \cos \theta)^2} \right) a(1 - \cos \theta)^2$$

$$= a\theta - a\sin \theta + \frac{a\sin \theta}{(1 - \cos \theta)} [1 + \cos^2 \theta - 2\cos \theta + \sin^2 \theta]$$

$$= a\theta - a\sin \theta + \frac{a\sin \theta}{(1 - \cos \theta)} (2 - 2\cos \theta)$$

$$= a\theta - a\sin \theta + \frac{2a\sin \theta(1 - \cos \theta)}{(1 - \cos \theta)}$$

$$\alpha = a\theta + a\sin \theta = a(\theta + \sin \theta) \quad \text{--- (I)}$$

$$\beta = \gamma + \frac{(1 + \gamma'^2)}{\gamma''} = a(1 - \cos \theta) + \left[\frac{1 + \sin^2 \theta}{(1 - \cos \theta)^2} \right] \frac{-1}{a(1 - \cos \theta)}$$

$$= a(1 - \cos \theta) - \left[\frac{(1 - \cos \theta)^2 + \sin^2 \theta}{(1 - \cos \theta)^2} \right] a(1 - \cos \theta)^2$$

$$= a(1 - \cos \theta) - a[1 + \cos^2 \theta - 2\cos \theta + \sin^2 \theta]$$

$$= a(1 - \cos \theta) - a(1 + 1 - 2\cos \theta) = a(1 - \cos \theta) - 2a(1 - \cos \theta)$$

$$\beta = (1 - \cos \theta)[a - 2a] = -a(1 - \cos \theta) \quad \text{--- (II)}$$

Elimination of θ is not possible. (I) & (II) are parametric Eqs of evolute, which is cycloid.

7.8-14

Q11.7 $x = a \cos^3 \theta$ $y = a \sin^3 \theta$

$$\frac{dx}{d\theta} = 3a \cos^2 \theta (-\sin \theta)$$

$$\frac{dy}{d\theta} = 3a \sin^2 \theta \cos \theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx} = \frac{3a \sin^2 \theta \cos \theta}{-3a \cos^2 \theta \sin \theta} = -\tan \theta$$

$$y'' = -\sec^2 \theta \frac{d\theta}{dx} = -\sec^2 \theta \frac{1}{3a \cos^2 \theta \sin \theta} = \frac{1}{3a \cos^4 \theta \sin \theta}$$

$$\alpha = x - y' \left(\frac{1+y'^2}{y''} \right)$$

$$= a \cos^3 \theta - (-\tan \theta) \left(\frac{1 + \tan^2 \theta}{\frac{1}{3a \cos^4 \theta \sin \theta}} \right) = a \cos^3 \theta + \frac{\sin \theta (\sec^2 \theta) 3a \cos^4 \theta \sin \theta}{\cos \theta}$$

$$= a \cos^3 \theta + \left(\frac{\sin \theta}{\cos \theta} \right) \left(\frac{1}{\cos^2 \theta} \right) 3a \cos^4 \theta \sin \theta$$

$$\alpha = a \cos^3 \theta + 3a \cos \theta \sin^2 \theta \quad \text{①}$$

$$\beta = y + \frac{1+y'^2}{y''}$$

$$= a \sin^3 \theta + \left(\frac{1 + \tan^2 \theta}{\frac{1}{3a \cos^4 \theta \sin \theta}} \right) = a \sin^3 \theta + (\sec^2 \theta) (3a \cos^4 \theta \sin \theta)$$

$$= a \sin^3 \theta + \frac{1}{\cos^2 \theta} 3a \cos^4 \theta \sin \theta$$

$$= a \sin^3 \theta + 3a \cos^2 \theta \sin \theta \quad \text{②}$$

7.8-15

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$$\begin{aligned}\alpha + \beta &= a \cos^3 \theta + 3a \sin^2 \theta \cos \theta + a \sin^3 \theta + 3a \cos^2 \theta \sin \theta \\ &= a (\cos^3 \theta + \sin^3 \theta + 3 \cos^2 \theta \sin \theta + 3 \sin^2 \theta \cos \theta)\end{aligned}$$

$$= a [\cos^3 \theta + \sin^3 \theta + 3 \cos \theta \sin \theta (\cos \theta + \sin \theta)]$$

$$\alpha + \beta = a (\cos \theta + \sin \theta)^3$$

$$(\alpha + \beta)^{\frac{2}{3}} = a^{\frac{2}{3}} \left[(\cos \theta + \sin \theta)^3 \right]^{\frac{2}{3}} = a^{\frac{2}{3}} (\cos \theta + \sin \theta)^2 \quad \text{--- (III)}$$

Now consider

$$\alpha - \beta = a \cos^3 \theta + 3a \sin^2 \theta \cos \theta - a \sin^3 \theta - 3a \cos^2 \theta \sin \theta$$

$$= a [\cos^3 \theta - \sin^3 \theta - 3 \cos^2 \theta \sin \theta + 3 \sin^2 \theta \cos \theta]$$

$$= a [\cos^3 \theta - \sin^3 \theta - 3 \cos \theta \sin \theta (\cos \theta - \sin \theta)]$$

$$\alpha - \beta = a (\cos \theta - \sin \theta)^3$$

$$(\alpha - \beta)^{\frac{2}{3}} = a^{\frac{2}{3}} \left[(\cos \theta - \sin \theta)^3 \right]^{\frac{2}{3}} = a^{\frac{2}{3}} (\cos \theta - \sin \theta)^2 \quad \text{--- (IV)}$$

Adding (III) & (IV) $(\alpha + \beta)^{\frac{2}{3}} + (\alpha - \beta)^{\frac{2}{3}} = a^{\frac{2}{3}} (\cos \theta + \sin \theta)^2 + a^{\frac{2}{3}} (\cos \theta - \sin \theta)^2$

$$= a^{\frac{2}{3}} \left[\cos^2 \theta + \sin^2 \theta + 2 \sin \theta \cos \theta + \cos^2 \theta + \sin^2 \theta - 2 \sin \theta \cos \theta \right]$$

$$= a^{\frac{2}{3}} [1 + 1]$$

$$(\alpha + \beta)^{\frac{2}{3}} + (\alpha - \beta)^{\frac{2}{3}} = 2a^{\frac{2}{3}}$$

Replace α by x & β by y

$$(x + y)^{\frac{2}{3}} + (x - y)^{\frac{2}{3}} = 2a^{\frac{2}{3}}$$

7.8-16

$$= a \left[\cos t + \ln \tan \frac{t}{2} \right] - \frac{1}{\cos t} \cdot \frac{1}{\cos^3 t} a \cos^4 t$$

$$= a \cos t + a \ln \tan \frac{t}{2} - a \cos t$$

$$\alpha = a \ln \tan \frac{t}{2}$$

①

$$\beta = y + \int \frac{1+y'^2}{y''}$$

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$$= a \sin t + \frac{(1 + \tan^2 t)}{\frac{\sin t}{a \cos^4 t}} = a \sin t + (\sec^2 t) \frac{a \cos^4 t}{\sin t}$$

$$= a \sin t + \frac{1}{\cos^2 t} \frac{a \cos^4 t}{\sin t} = a \sin t + \frac{a \cos^2 t}{\sin t}$$

$$\beta = \frac{a \sin^2 t + a \cos^2 t}{\sin t} = \frac{a (\sin^2 t + \cos^2 t)}{\sin t} = \frac{a}{\sin t}$$

$$\beta = \frac{a}{2 \sin \frac{t}{2} \cos \frac{t}{2}} = \frac{\frac{a}{\cos^2 \frac{t}{2}}}{2 \sin \frac{t}{2} \cos \frac{t}{2}} = \frac{a \sec^2 \frac{t}{2}}{2 \tan \frac{t}{2}}$$

$$\beta = \frac{a (1 + \tan^2 \frac{t}{2})}{2 \tan \frac{t}{2}} \quad \text{--- ②}$$

For evolute eliminate t from ① & ②

$$\text{As } \alpha = a \ln \tan \frac{t}{2} \quad \frac{\alpha}{a} = \ln \tan \frac{t}{2}$$

$$e^{\frac{\alpha}{a}} = \tan \frac{t}{2}$$

$$(x-y) + (x-y)' = a' \alpha$$

7.8-17

Q8 $x = a \left(\cos t + \ln \tan \left(\frac{t}{2} \right) \right)$ $y = a \sin t$

$$\frac{dx}{dt} = a \left(-\sin t + \frac{1}{\tan \left(\frac{t}{2} \right)} \sec^2 \left(\frac{t}{2} \right) \frac{1}{2} \right) \quad \frac{dy}{dt} = a \cos t$$

$$= a \left(-\sin t + \frac{1}{\cos^2 \left(\frac{t}{2} \right) \sin \left(\frac{t}{2} \right)} \frac{1}{2} \right)$$

$$= a \left(-\sin t + \frac{1}{2 \cos^2 \frac{t}{2} \sin \frac{t}{2}} \right)$$

$$= a \left(-\sin t + \frac{1}{\sin t} \right) = a \left(\frac{-\sin^2 t + 1}{\sin t} \right)$$

$$\frac{dx}{dt} = a \frac{\cos^2 t}{\sin t}$$

$$y' = \frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{a \cos t}{a \frac{\cos^2 t}{\sin t}} = \frac{\sin t}{\cos t} = \tan t$$

$$y'' = \sec^2 t \frac{dt}{dx} = \sec^2 t \frac{\sin t}{a \cos^2 t} = \frac{\sin t}{a \cos^4 t}$$

(d) Centre of curvature (2, 3)

$$\alpha = x - y' \left(\frac{1 + y'^2}{y''} \right)$$

$$= a \left(\cos t + \ln \tan \left(\frac{t}{2} \right) \right) - \tan t \left(\frac{1 + \tan^2 t}{\frac{\sin t}{a \cos^4 t}} \right)$$

$$= a \left(\cos t + \ln \tan \frac{t}{2} \right) - \frac{\sin t}{\cos t} \left(\sec^2 t \right) \frac{a \cos^4 t}{\sin t}$$

7.8-18

$$\therefore \textcircled{III} \text{ becomes } \beta = a \frac{(1 + e^{\frac{2\alpha}{a}})}{2e^{\frac{\alpha}{a}}}$$

$$\beta = \frac{a}{2} \left(\frac{1}{e^{\frac{\alpha}{a}}} + \frac{e^{\frac{2\alpha}{a}}}{e^{\frac{\alpha}{a}}} \right)$$

$$= \frac{a}{2} \left(e^{-\frac{\alpha}{a}} + e^{\frac{\alpha}{a}} \right) = a \left(\frac{e^{-\frac{\alpha}{a}} + e^{\frac{\alpha}{a}}}{2} \right)$$

$$\beta = a \cosh \frac{\alpha}{a}$$

- Replacing α by x & β by y

$$y = a \cosh \frac{x}{a}$$

Def $x^3 + y^3 = 3axy$.

Centre of curvature at $(\frac{3a}{2}, \frac{3a}{2})$ is $(\frac{21a}{16}, \frac{21a}{16})$

$$3x^2 + 3y^2 \frac{dy}{dx} = 3a(x \frac{dy}{dx} + y)$$

$$3y^2 \frac{dy}{dx} - 3ax \frac{dy}{dx} = 3ay - 3x^2$$

$$3 \frac{dy}{dx} (y^2 - ax) = 3(ay - x^2)$$

$$\frac{dy}{dx} = \frac{3(ay - x^2)}{3(y^2 - ax)} \quad \text{--- (1)}$$

$$\text{At } (\frac{3a}{2}, \frac{3a}{2}) \quad \frac{dy}{dx} = \frac{a(\frac{3a}{2}) - (\frac{3a}{2})^2}{(\frac{3a}{2})^2 - a(\frac{3a}{2})} = \frac{\frac{3a^2}{2} - \frac{9a^2}{4}}{\frac{9a^2}{4} - \frac{3a^2}{2}}$$

7.8-19

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$$\frac{dy}{dx} = -\left(\frac{9a^2}{4} - \frac{3a^2}{2}\right) = -1$$

from ①

$$\frac{d^2y}{dx^2} = \frac{1}{(y^2 - ax)^2} ((y^2 - ax)(ay' - 2x) - (ay - x^2)(2yy' - a))$$

$$\text{At } \left(\frac{3a}{2}, \frac{3a}{2}\right) \frac{d^2y}{dx^2} = \left(\frac{9a^2}{4} - \frac{3a^2}{2}\right)(a(-1) - 2\left(\frac{3a}{2}\right)) - \left(\frac{3a^2}{2} - \frac{9a^2}{4}\right)(2\left(\frac{3a}{2}\right)(-1) - a)$$

$$= \frac{\left(\frac{9a^2}{4} - \frac{3a^2}{2}\right)^2}{\left(\frac{9a^2}{4} - \frac{3a^2}{2}\right)^2} \left[(-a - 3a) + (-3a - a)\right]$$

$$\left(\frac{9a^2}{4} - \frac{3a^2}{2}\right)^2 =$$

$$= \frac{-8a}{\frac{9a^2}{4} - \frac{3a^2}{2}} = \frac{-8a}{\frac{9a^2 - 6a^2}{4}} = \frac{-32a}{3a^2}$$

$$y'' = \frac{-32}{3a}$$

$$\alpha = x - y' \left(\frac{1+y'^2}{y''} \right) = \frac{3a}{2} - (-1) \left(\frac{1+(-1)^2}{\frac{-32}{3a}} \right)$$

$$= \frac{3a}{2} + 1 \frac{(1+1)}{\frac{-32}{3a}} 3a = \frac{3a}{2} - \frac{6a}{32} = \frac{3a}{2} - \frac{3a}{16} = \frac{24a - 3a}{16} = \frac{21a}{16}$$

Similarly

$$\beta = y + \left(\frac{1+y'^2}{y''} \right)$$

$$= \frac{3a}{2} + \frac{(1+1)}{\frac{-32}{3a}} = \frac{3a}{2} - \frac{6a}{32}$$

$$\beta = \frac{3a}{2} - \frac{6a}{32} = \frac{24a - 3a}{16} = \frac{21a}{16}$$

Q10 Let $P(x, y)$ be any pt on a curve $y = f(x)$

We know by Theorem that coord of the centre of curvature $C(\alpha, \beta)$ corresponding to the pt P are

$$\alpha = x - \rho \sin \psi \quad \text{--- (1)}$$

$$\beta = y + \rho \cos \psi \quad \text{--- (2)}$$

Diff (1) & (2) w.r.t 'x'

$$\frac{d\alpha}{dx} = 1 - \left[\rho \cos \psi \frac{d\psi}{dx} + \sin \psi \frac{d\rho}{dx} \right]$$

$$= 1 - \left[\frac{ds}{dy} \cdot \frac{dx}{ds} \cdot \frac{d\psi}{dx} + \sin \psi \frac{d\rho}{dx} \right]$$

$$= 1 - \frac{dx}{ds} \cdot \frac{ds}{dy} \cdot \frac{d\psi}{dx} - \sin \psi \frac{d\rho}{dx}$$

$$= 1 - 1 - \sin \psi \frac{d\rho}{dx}$$

$$\frac{d\alpha}{dx} = - \sin \psi \frac{d\rho}{dx} \quad \text{--- (3)}$$

$$\frac{d\beta}{dx} = \frac{dy}{dx} - \rho \sin \psi \frac{d\psi}{dx} + \frac{d\rho}{dx} \cos \psi$$

$$= \frac{dy}{dx} - \frac{ds}{dy} \cdot \frac{dy}{ds} \cdot \frac{d\psi}{dx} + \frac{d\rho}{dx} \cos \psi$$

$$\frac{d\beta}{dx} = \frac{dy}{dx} - \frac{dy}{dx} + \frac{d\rho}{dx} \cos \psi$$

$$= \frac{d\rho}{dx} \cos \psi \quad \text{--- (4)}$$

7.8-21

$$\frac{d\beta}{d\alpha} = \frac{d\beta/dx}{d\alpha/dx} = \frac{\frac{dP}{dx} \cos \psi}{\frac{dP}{dx} (-\sin \psi)}$$

$$\frac{d\beta}{d\alpha} = -\cot \psi = -\frac{1}{\tan \psi} = -\frac{1}{\frac{dy}{dx}} \left(\frac{\text{slope of Normal}}{\text{slope of Tangent}} \right) \quad \left(\text{like } \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} \right)$$

Now $\frac{d\beta}{d\alpha}$ is slope of the tangent to the evolute at $C(\alpha, \beta)$ and it is equal to the slope of the normal PC to the curve $y = f(x)$ at $P(x, y)$.

